Journal of The
Electrochemical Society

OPEN ACCESS

Accurate LLI and LAM_{PE} Estimation Using the Mechanistic Modeling Approach with Layered Oxides

To cite this article: Matthieu Dubarry *et al* 2023 *J. Electrochem. Soc.* **170** 070503

View the article online for updates and enhancements.

You may also like

- A Simulation Framework for Electrochemical Processes with Electrolyte Flow
Mengyuan Huang, Norbert Weber and Gerd Mutschke

- Corrosion Electrochemistry of Chromium in Molten FLiNaK Salt at 600 °C
Ho Lun Chan, Elena Romanovskaia, Valentin Romanovski et al.

- Warburg Conductivity for Binary Electrolyte for Enabling Electrolyte Screening and Efficient Battery Operations
Basudev Choudhury, Akash Jangale and Bharatkumar Suthar

Your Lab in a Box!

EL-CELL®
electrochemical test equipment

The PAT-Tester-i-16 Multi-Channel Potentiostat for Battery Material Testing!

- All-in-One Solution with Integrated Temperature Chamber (+10 to +80 °C)!
No additional devices are required to measure at a stable ambient temperature.
- Fully Featured Multi-Channel Potentiostat / Galvanostat / EIS!
Up to 16 independent battery test channels, no multiplexing.
- Ideally Suited for High-Precision Coulometry!
Measure with excellent accuracy and signal-to-noise ratio.
- Small Footprint, Easy to Setup and Operate!
Cableless connection of 3-electrode battery test cells. Powerful EL-Software included.

Learn more on our product website: Scan me!

Download the data sheet (PDF): Scan me!

Or contact us directly:
+49 40 79012-734
sales@el-cell.com
www.el-cell.com

This content was downloaded from IP address 216.209.51.241 on 18/03/2026 at 00:46

Journal of The Electrochemical Society, 2023 **170** 070503

Accurate LLI and LAM_{PE} Estimation Using the Mechanistic Modeling Approach with Layered Oxides

Matthieu Dubarry,[1,*,z] Vishal Agrawal,[2] Martin Hüske,[2] and Matthias Kuipers[2]

[1]*Hawai'i Natural Energy Institute, University of Hawai'i at Mānoa, Honolulu, Hawaii, United States of America*
[2]*Accure Battery Intelligence GmbH, Aachen, Germany*

In the past decade and half since their introduction, multiple frameworks of mechanistic models were made available to enable the quantification of loss of lithium inventory and loss of active materials. They usually come either with capacity/state of charge- or lithiation-based matching for the electrodes. While both approaches were believed to be equivalent, the present work will show that they are not when used with materials that are not fully delithiated at end-of-charge because they differ on how they account for inaccessible lithium. Herein, we will explain these discrepancies and propose new equations to improve both types of frameworks.
© 2023 The Author(s). Published on behalf of The Electrochemical Society by IOP Publishing Limited. This is an open access article distributed under the terms of the Creative Commons Attribution 4.0 License (CC BY, http://creativecommons.org/licenses/by/4.0/), which permits unrestricted reuse of the work in any medium, provided the original work is properly cited. [DOI: 10.1149/1945-7111/ace21c]

Manuscript submitted March 1, 2023; revised manuscript received April 21, 2023. Published July 5, 2023.

The mechanistic modeling approach has attracted significant interest in recent years because of its ability to emulate the impact of degradation modes, i.e., loss of lithium inventory (LLI), loss of active material (LAM) on the positive and negative electrodes (PE and NE respectively), as well as kinetic limitations.[1] Since the seminal work by Bloom et al.,[2] a multitude of frameworks were published by different groups[2–37] using the same basic principles of matching the voltage response of PE and NE half-cells to emulate the response of a full cell (FC). The different frameworks can mainly be classified into two categories depending on what parameters are used for the electrode matching: capacity (Q)/state of charge (SOC) or lithiation. Most of the pioneering models[2–7] were Q/SOC-based and relied on at minimum two parameters to match the electrodes, the ratio of capacity between PE and NE (LR) and their offset or slippage (OFS). The lithiation-based models[23–34,38,39] relied on a minimum of four parameters, the minimum and maximum lithiation operation values for each electrode. To best of our knowledge, both approaches have always been considered equivalent. We used and kept using a Q/SOC-based approach with the 'alawa toolbox[5] because, with less parameters to optimize or scan, it is faster for multiple calculations, such as synthetic data generation.[40,41]

While both approaches are equivalent when used with materials using 100% of their lithium, this is not the case for materials which do not because of the presence of inaccessible lithium.[38,39,42] This will affect lithium cobalt oxide (LCO), nickel cobalt aluminum oxide (NCA), and nickel manganese cobalt oxide (NMC) that typically display lithiations between 0.33 and 0.5 in the 4.2 to 4.3 V window. This work complements previous descriptions of the issue[38,39] to illustrate that the usage of only a portion of the available lithium ions induces different issues in both categories of mechanistic models in their current implementations. The Q/SOC-based models are able to correctly quantify and simulate the voltage response induced by LAM_{PE} but are overestimating LLI. The lithiation-based models that are not considering the impact of inaccessible lithium properly might not properly quantify LAM_{PE}, not accurately simulate its voltage response and overestimate LLI. This is because LAM_{PE} typically refers to the loss of an entirely delithiated electrode. However, throughout the usage of the cell, the electrode will never be fully delithiated, and therefore a fully delithiated LAM_{PE} cannot happen. At minimum, LAM_{PE} will always include the lithium ions that are left at the cut-off voltage.

The present work aims at illustrating and explaining the origin of these inaccuracies to allow for the models to account for this more accurate representation of the true size of the PE without having to use half-cell data covering the entire delithiation process. While this issue might appear inconsequential and to be induced from different definitions and assumptions, it is not the case as comparing the evolution of the ratio of LLI to LAMs is a good indicator of a possible acceleration of the capacity loss,[43,44] therefore an ill-quantification might have significant safety repercussions.

Impact Of Considering Full Delithiation on LAM_{PE} and LLI

Looking into the literature, the lithiation extent x_{cutoff} can be estimated for typical layered oxide battery materials. For NMC111, lithiations of 0.42 @ 4.2 V and 0.33 @ 4.3 V can be estimated from Ref. 45. For LCO, lithiation can be estimated at around 0.50 @ 4.2 V and 0.45 @ 4.3 V.[46] Full lithiation range for NCA can be found in Ref. 47 for a C/10 charge from which it can be estimated that the lithiation is likely around 0.4 @ 4.2 V and 0.35 @ 4.3 V for the open circuit voltage. Finally, NMC811 showcases little capacity above 4.4 V,[48] similar to what was reported for $LiNiO_2$.[49] The values used in previously published mechanistic models are variable (0.38 @ 4.25 V,[23] 0.5 @ 4.3 V,[24] 0.2 @ 4.4 V, and 0.0 @ 4.5 V[26–30] for NMC111, 0.4 @ 4.4 V for LCO[34]) often without justification nor any references provided.

The Real size of the electrode is important to visualize the true impact of LAM_{PE}. Figure 1 illustrates the impact of full delithiation on the emulation of different amount of LAM_{PE} (0% in blue, 10% in red, and 20% in green) for an exemplary NMC111 PE for a C/25 charge vs metallic lithium. Two sets of curves are plotted for the PEs, the full lines represent traditional half-cell data tested up to a 4.3 V cut-off voltage and the dashed lines represent the response of the same material synthetically extended by adding 50% capacity between 4.3 V and 5.0 V to represent what a fully delithiated electrode would look like with a x_{cutoff} around 0.33 @ 4.3 V.[45]

LAM_{PE} induces some shrinkage of the PE[5] and its impact is mostly visible at low capacity where the lower bound of the PE capacity normalized to the initial one will increase to the level of LAM_{PE}. For both sets to have the same impact on the minimum PE capacity, the actual amount of LAM_{PE} is lower when the fully delithiated curves are used because the considered electrode range is larger. This is also true for LLI because the percentage of consumed Li-ions depends on the size of the PE[50] as the only lithium available is the one that was provided by the initial PE. The other impact of LAM_{PE} on the PE response is visible at higher voltages. Because the real LAM_{PE} is lower when the fully delithiated curves are used, the PE end-of-charge (EOC) state exceeds the initial 100% at the cutoff voltage (dashed black lines on Fig. 1). E.g., for 20% LAM_{PE}, the minimum capacity will still be of 20% but the PE will be reduced by 13.33% instead of 20% if considering the full delithiation (20*100/(100 + 50) = 13.33). This implies that the capacity of the electrode

*Electrochemical Society Member.
zE-mail: matthieu@hawaii.edu

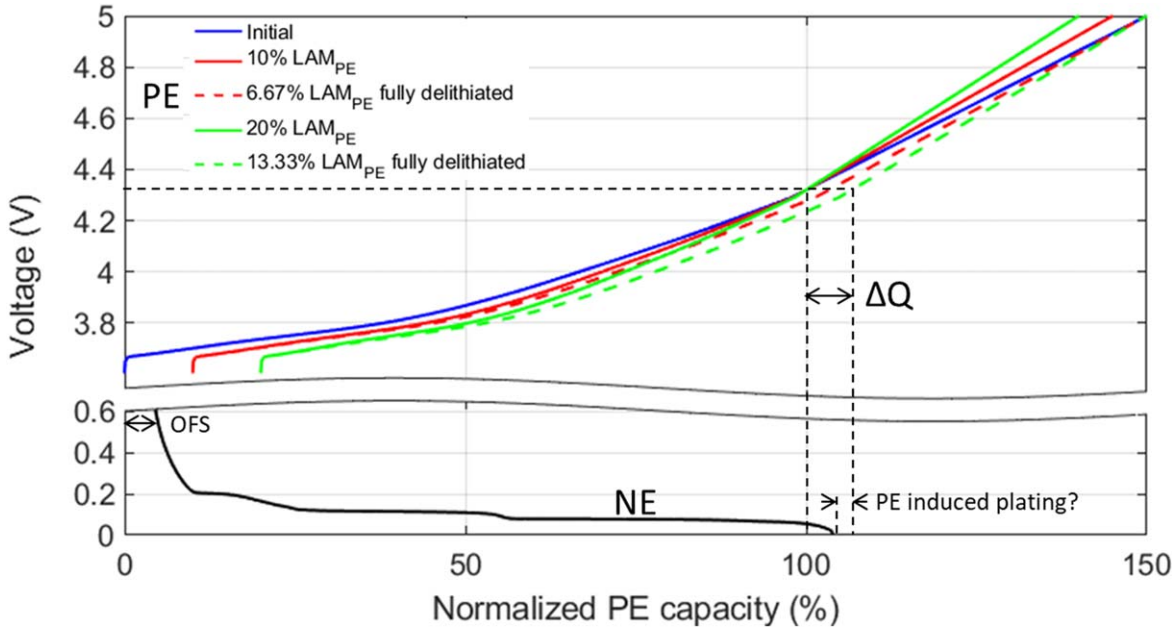

Figure 1. Effect of up to 20% LAM_{PE} on a C/25 charge for an exemplary NMC111 and graphite electrodes vs metallic lithium for cutoffs at 4.3 V and 5.0 V using the initial capacity as reference.

is now 86.67% and adding the 20% minimum capacity brings the maximum capacity to 106.67% at 4.3 V. This PE capacity excess ΔQ can be quantified as $x_{cutoff}.LAM_{PE}$ which corresponds to 3.33% and 6.67% for 10% and 20% LAM_{PE} respectively in this example.

With a PE not fully delithiated at cutoff, this capacity excess could have two major implications at the FC level where the NE also needs to be considered. First, the FC capacity could initially increase with LAM_{PE}. Indeed, while the PE capacity gain induced by ΔQ is lower than the PE capacity loss, resulting in a lower overall PE capacity, this is not the case at the FC level as the PE is usually not limiting at end of discharge.[50] Until the PE becomes the limiting electrode at end of discharge (i.e. $LAM_{PE} > OFS$), there is no FC capacity loss induced by LAM_{PE} to counterbalance the excess. Therefore, there could be a capacity gain. Second, it represents a potential pathway leading to PE induced lithium plating[44] because the usage of PE capacity over 100% implies an increased lithiation of the NE on the other side. As a result, the plating threshold[51] could be drastically reduced and plating could be gradually initiated by LAM_{PE} with a full NE before all the lithium are released from the PE as exemplified in Fig. 1. This could be especially true at higher rates where the rate capability of graphite might be lowered which will decrease the NE full lithiation SOC.

Whether this could happen or not is where both current implementations of the mechanistic model diverge. A typical lithiation-based model would predict that both would happen while a Q/SOC-based model won't. The discrepancy between the models is induced by the way LAM_{PE} is handled. In Q/SOC-based models, LAM_{PE} refers to the extent of PE that is useable for reaction (assuming the cutoff of the experimental half-cell data is close to the one of the FC plus ∼100 mV to account for the graphite NE). For a lithiation-based model it does not, and LAM_{PE} refers to the loss of an entirely delithiated electrode. However, the latter cannot happen because LAM_{PE} will always include at least the lithium ions that are left at the cut-off voltage. Accounting for this inaccessible lithium lost in the active material will negate the impact of excess capacity and avoid the possibility of plating, reconciling both model categories.

Model Accommodation

Equations for the correct accommodation for lithiation-based models are already available in the literature.[38,39] The following section is more focused on a mechanistic discussion of the impact of the electrode size. Before getting into the details, it is important to first discuss what SOC scale to use to avoid any possible confusion. Two SOC scales could be used, first the "usable" SOC, i.e., the capacity that the electrode can actually use to the selected cut-off and second the true SOC, SOC_{true}, that considers the full lithiation range. While the latter is more physical, it will also be less intuitive (LR«1, harder to compare to other work) and will necessitate a full rescaling of the PE half-cell data which will cost some calculation time for large synthetic datasets.[40,41] Therefore, we elected to keep 100% SOC as 100% of the capacity of the PE half-cell data up to the experimental cutoff voltage but we will differentiate between the true LAM_{PE} and true LLI as calculated from the full delithiated electrode, and the usable ones. In addition, since some lithium is lost inside the active material, the lithiation of the LAM_{PE}^{True} also needs to be considered. When lithium is present in the AM, and to ease discussions, the LAM can be decomposed in its fully lithiated (li-) and fully delithiated (de-) components:[5]

$$LAM_{PE} = LAM_{dePE} + LAM_{liPE} \quad [1]$$

LAM_{liPE} is rarely discussed in the literature because it is generally impossible to quantify with certitude where the Lithium was actually lost (in material vs passivated) without a specific set of circumstances (e.g.[52]). Therefore this lithium is usually quantified as LLI with LAM_{PE} always considered fully delithiated[50,53] with $LAM_{liPE} = 0$. However, in the case where full delithiation is used, LAM_{liPE}^{True} can be quantified because the material is lost with a known amount of lithium inside, $LAM_{x_{cutoff}PE}^{True}$. Assuming $LAM_{x_{cutoff}PE}^{True} = 1$, LAM_{liPE}^{True} must then be equal to $x_{cutoff.}$ to bring the necessary lithium and, using Eq. 1, LAM_{dePE}^{True} must be equal to $(1 - x_{cutoff})$ to complement. This allows to express the relationship between LAM_{liPE}^{True}, LAM_{dePE}^{True}, and x_{cutoff}:

$$LAM_{liPE}^{True} = \frac{x_{cutoff}}{(1 - x_{cutoff})} . LAM_{dePE}^{True} \quad [2]$$

With this relationship established, the true and usable scales can be related. Since the shrinkage of the PE at lower capacities is solely induced by LAM_{dePE},[5] and to induce the same shift when the full and the usable PE are used, LAM_{dePE}^{True} has to be x_{cutoff} less than

$LAM_{\mathrm{dePE}}^{\mathrm{Usable}}$ because of the different electrode sizes (Fig. 1):

$$LAM_{\mathrm{dePE}}^{\mathrm{True}} = (1 - x_{\mathrm{cutoff}}) \,.\, LAM_{\mathrm{dePE}}^{\mathrm{Usable}} \quad [3]$$

$LAM_{\mathrm{liPE}}^{\mathrm{True}}$ can then be expressed as a function of . using Eq. 2:

$$LAM_{\mathrm{liPE}}^{\mathrm{True}} = x_{\mathrm{cutoff}} \,.\, LAM_{\mathrm{dePE}}^{\mathrm{Usable}} \quad [4]$$

Eqs. 1, 3 and 4 can then relate the total amount of $LAM_{\mathrm{PE}}^{\mathrm{True}}$ to $LAM_{\mathrm{dePE}}^{\mathrm{Usable}}$. This can be extended to $\mathrm{LAM}_{\mathrm{PE}}^{\mathrm{Usable}}$ since any additional lithium lost in the active material cannot be formerly quantified and must thus be accounted as LLI with $LAM_{\mathrm{liPE}}^{\mathrm{Usable}}$ equal to 0:

$$LAM_{\mathrm{PE}}^{\mathrm{True}} = (1 - x_{\mathrm{cutoff}}) \,.\, LAM_{\mathrm{dePE}}^{\mathrm{Usable}} + x_{\mathrm{cutoff}} \,.\, LAM_{\mathrm{dePE}}^{\mathrm{Usable}} = LAM_{\mathrm{dePE}}^{\mathrm{Usable}} = LAM_{\mathrm{PE}}^{\mathrm{Usable}} \quad [5]$$

This proves that the total $\mathrm{LAM}_{\mathrm{PE}}$ is the same independently of the lithiation range considered. This relationship is illustrated schematically on Fig. 2 where, in case of fully delithiated electrode, the voltage response needs to be shrank from both sides to account for the true lithiated and delithiated losses that are happening concurrently. Because of the fixed ratio between the two, Eq. 2, and independently of the actual amount of $\mathrm{LAM}_{\mathrm{PE}}$, the shrank electrode will always cross the pristine one at the chosen cutoff voltage. This will negate any risk of excess capacity and will always overlap the electrode response after $\mathrm{LAM}_{\mathrm{dePE}}^{\mathrm{Usable}}$ that uses the cutoff voltage as a reference.

Summarizing, Q/SOC-based models that are using a narrow lithiation range do not account for the lithium lost in the inaccessible material, but this does not change the voltage response of the cell in the usable voltage window. Therefore, these models accurately quantify and simulate the impact of the total $\mathrm{LAM}_{\mathrm{PE}}$. However, what they assume is only $\mathrm{LAM}_{\mathrm{dePE}}$ is in reality $\mathrm{LAM}_{\mathrm{xcutoffPE}}$ if the full electrode is considered. In addition, LLI is also proportional to the size of the electrode and the latter should therefore be specified when reporting results. Models using the full lithiation range must take the lithiation extent of $\mathrm{LAM}_{\mathrm{PE}}$ into account to avoid wrongly predicting the SOC shift mentioned above while underestimating the total $\mathrm{LAM}_{\mathrm{PE}}$ by only accounting for $\mathrm{LAM}_{\mathrm{dePE}}$ and overestimating LLI by ignoring $\mathrm{LAM}_{\mathrm{liPE}}$.

It has to be noted that, for blends the $\mathrm{LAM}_{\mathrm{PE}}$ would need to be separated in component specific values where the specific x_{cutoff} could be different, such as 0.33 and 0 respectively for an NMC/$LiMn_2O_4$ (LMO) blend at 4.3 V. Additionally, to simulate a gradual overcharge, x_{cutoff} would need to be updated. Finally, the lithiation at the cutoff was used instead of a lithiation range to prevent issues with materials that can be overlithiated while in blends like LMO.[54]

Conclusions

In this work we analyzed in detail the impact of the $\mathrm{LAM}_{\mathrm{PE}}$ for materials for which full delithiation is not achieved at end-of-charge. This includes some of the most common PE materials such as LCO, NCA, and NMC. This work showcases that, in case of pure delithiated $\mathrm{LAM}_{\mathrm{PE}}$ and because part of the electrode at high voltage is not used, the EOC SOC of the PE could be seen as increasing with $\mathrm{LAM}_{\mathrm{PE}}$ with some excess capacity available proportionally to the amount of material lost and the actual lithiation at selected cutoff voltage. While this would be of major concern for safety and potential $\mathrm{LAM}_{\mathrm{PE}}$ induced plating, this excess capacity is not achievable in real batteries because the PE is never fully delithiated and thus pure $\mathrm{LAM}_{\mathrm{PE}}$ is not possible. The least lithiated the PE could be is the lithiation at the cutoff voltage and this lithium lost in the PE will at least shift the NE enough to negate the possibility of excess capacity.

Looking at how the different types of mechanistic models could handle this case-figure, it was found that Q/SOC-based models were taking this problem into account by default because they relied on the available capacity only. However, they were overestimating LLI because only a part of the electrode's lithiation range was considered. Most lithiation-based models however would not generate the right voltage response for pure $\mathrm{LAM}_{\mathrm{PE}}$ because of the lack on consideration of the inaccessible lithium lost in the PE. This results in an underestimation of $\mathrm{LAM}_{\mathrm{PE}}$ and an overestimation of LLI as the contribution of the real LLI and the one in inaccessible material cannot be separated. The present paper proposed a theoretical background to reconcile both approaches in emulating $\mathrm{LAM}_{\mathrm{PE}}$ properly.

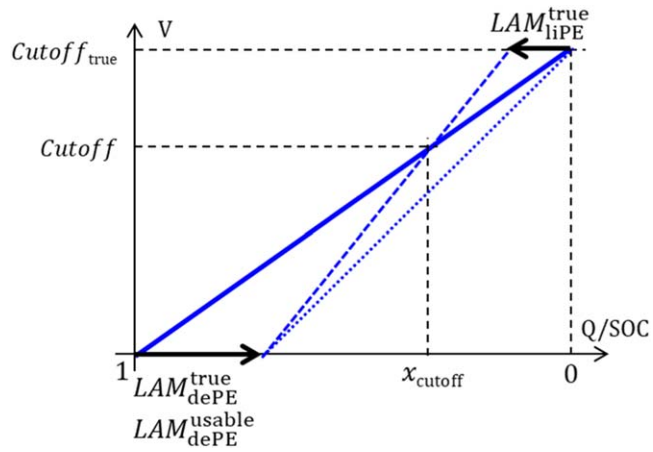

Figure 2. Schematic representation of the impact of $\mathrm{LAM}_{\mathrm{PE}}$ on both the usable and the true SOC scale for a cut-off voltage below the full-delithiation one.

Acknowledgments

This work was funded by the Office of Naval Research (ONR) grant number N00014–19–1–2159 and by ACCURE Battery Intelligence.

ORCID

Matthieu Dubarry https://orcid.org/0000-0002-3228-1834

References

1. M. Dubarry and D. Beck, *Accounts of Materials Research*, **3**, 843 (2022).
2. I. Bloom, A. N. Jansen, D. P. Abraham, J. Knuth, S. A. Jones, V. S. Battaglia, and G. L. Henriksen, *J. Power Sources*, **139**, 295 (2005).
3. K. Honkura, H. Honbo, Y. Koishikawa, and T. Horiba, *ECS Trans.*, **13**, 61 (2008).
4. K. Honkura, K. Takahashi, and T. Horiba, *J. Power Sources*, **196**, 10141 (2011).
5. M. Dubarry, C. Truchot, and B. Y. Liaw, *J. Power Sources*, **219**, 204 (2012).
6. H. M. Dahn, A. J. Smith, J. C. Burns, D. A. Stevens, and J. R. Dahn, *J. Electrochem. Soc.*, **159**, A1405 (2012).
7. R. Fathi, J. C. Burns, D. A. Stevens, H. Ye, C. Hu, G. Jain, E. Scott, C. Schmidt, and J. R. Dahn, *J. Electrochem. Soc.*, **161**, A1572 (2014).
8. J. P. Schmidt, H. Y. Tran, J. Richter, E. Ivers-Tiffée, and M. Wohlfahrt-Mehrens, *J. Power Sources*, **239**, 696 (2013).
9. J. Schmitt, M. Schindler, A. Oberbauer, and A. Jossen, *J. Power Sources*, **532**, 231296 (2022).
10. P. Keil, S. F. Schuster, J. Wilhelm, J. Travi, A. Hauser, R. C. Karl, and A. Jossen, *J. Electrochem. Soc.*, **163**, A1872 (2016).
11. A. Marongiu, N. Nlandi, Y. Rong, and D. U. Sauer, *J. Power Sources*, **324**, 158 (2016).
12. S. Schindler and M. A. Danzer, *J. Power Sources*, **343**, 226 (2017).
13. S. Gantenbein, M. Schönleber, M. Weiss, and E. Ivers-Tiffée, *Sustainability*, **11**, 6697 (2019).
14. Y. Gao, S. Yang, J. Jiang, C. Zhang, W. Zhang, and X. Zhou, *J. Electrochem. Soc.*, **166**, A1623 (2019).
15. J. Sturm, A. Rheinfeld, I. Zilberman, F. B. Spingler, S. Kosch, F. Frie, and A. Jossen, *J. Power Sources*, **412**, 204 (2019).
16. T. R. Tanim et al., *J. Electrochem. Soc.*, **166**, A1926 (2019).
17. I. Zilberman, J. Sturm, and A. Jossen, *J. Power Sources*, **425**, 217 (2019).
18. X. Jia, C. Zhang, L. Y. Wang, L. Zhang, and W. Zhang, *J. Electrochem. Soc.*, **167**, 020503 (2020).
19. S. Lee and Y. Kim, *2020 American Control Conference*, Denver, CO, USA (2020).

20. J. Chen, M. N. Marlow, Q. Jiang, and B. Wu, *Journal of Energy Storage*, **45**, 103669 (2022).
21. H. Ruan, J. Chen, W. Ai, and B. Wu, *Energy and AI*, **9**, 100158 (2022).
22. G. Fan, D. Lu, M. S. Trimboli, G. L. Plett, C. Zhu, and X. Zhang, *J. Power Sources*, **557**, 232555 (2023).
23. H. Ge, T. Aoki, N. Ikeda, S. Suga, T. Isobe, Z. Li, Y. Tabuchi, and J. Zhang, *J. Electrochem. Soc.*, **164**, A1050 (2017).
24. T. Lu, Y. Luo, Y. Zhang, W. Luo, L. Yan, and J. Xie, *J. Electrochem. Soc.*, **164**, A775 (2017).
25. M. Lewerenz, G. Fuchs, L. Becker, and D. U. Sauer, *Journal of Energy Storage*, **18**, 149 (2018).
26. S. Lee, P. Mohtat, J. B. Siegel, A. Stefanopoulou, J.-W. Lee, and T.-K. Lee, *IEEE Trans. Ind. Inf.*, **1**, 3376 (2019).
27. S. Lee, J. B. Siegel, A. G. Stefanopoulou, J.-W. Lee, and T.-K. Lee, *J. Electrochem. Soc.*, **167**, 090531 (2020).
28. C. R. Birkl, E. McTurk, M. R. Roberts, P. G. Bruce, and D. A. Howey, *J. Electrochem. Soc.*, **162**, A2271 (2015).
29. C. R. Birkl, E. McTurk, S. Zekoll, F. H. Richter, M. R. Roberts, P. G. Bruce, and D. A. Howey, *J. Electrochem. Soc.*, **164**, A2644 (2017).
30. C. R. Birkl, M. R. Roberts, E. McTurk, P. G. Bruce, and D. A. Howey, *J. Power Sources*, **341**, 373 (2017).
31. X. Han, M. Ouyang, L. Lu, J. Li, Y. Zheng, and Z. Li, *J. Power Sources*, **251**, 38 (2014).
32. S. Lee, P. Mohtat, J. B. Siegel, and A. G. Stefanopoulou, *2018 Annual American Control Conference (ACC), Wisconsin Center, Milwaukee, USA* (2018).
33. K. M. Mbeya, N. Damay, G. Friedrich, C. Forgez, and M. Juston, *Mathematics and Computers in Simulation*, **183**, 34 (2020).
34. J. Z. Olson, C. M. López, and E. J. F. Dickinson, *Chem. Mater.*, **35**, 1487 (2023).
35. P. Mohtat, S. Lee, J. B. Siegel, and A. G. Stefanopoulou, *J. Power Sources*, **427**, 101 (2019).
36. G. Liu, M. Ouyang, L. Lu, J. Li, and X. Han, *J. Power Sources*, **274**, 971 (2015).
37. A. Rahman and X. Lin, *Appl. Energy*, **321**, 119367 (2022).
38. D. Lu, M. Scott Trimboli, G. Fan, R. Zhang, and G. L. Plett, *J. Electrochem. Soc.*, **168**, 070532 (2021).
39. A. Weng, J. B. Siegel, and A. Stefanopoulou, *Frontiers in Energy Research*, **11** (2023).
40. M. Dubarry and D. Beck, *J. Power Sources*, **479**, 228806 (2020).
41. M. Dubarry and D. Beck, *Energies*, **14**, 2371 (2021).
42. C. Truchot, M. Dubarry, and B. Y. Liaw, *Appl. Energy*, **119**, 218 (2014).
43. M. Dubarry, G. Baure, and A. Devie, *J. Electrochem. Soc.*, **165**, A773 (2018).
44. P. M. Attia et al., *J. Electrochem. Soc.*, **169**, 0605517 (2022).
45. N. Yabuuchi, Y. Makimura, and T. Ohzuku, *Journal of The Electrochemical Society,*, **154**, A314 (2007).
46. D. Carlier, I. Saadoune, M. Ménétrier, and C. Delmas, *J. Electrochem. Soc.*, **149**, A1310 (2002).
47. W.-S. Yoon, K. Y. Chung, J. McBreen, D. A. Fischer, and X.-Q. Yang, *J. Power Sources*, **174**, 1015 (2007).
48. B. Huang, M. Wang, Z. Zhao, L. Chen, and Y. Gu, *J. Alloys Compd.*, **810**, 151800 (2019).
49. C. Delmas, M. Menetrier, L. Croguennec, S. Levasseur, J. P. Peres, C. Pouillerie, G. Prado, L. Fournes, and F. Weill, *Int. J. Inorg. Mater.*, **1**, 11 (1999).
50. M. Dubarry and D. Anseán, *Frontiers in Energy Research*, **10** (2022).
51. G. Baure and M. Dubarry, *Batteries*, **5**, 42 (2019).
52. A. Devie, M. Dubarry, H.-P. Wu, T.-H. Wu, and B. Y. Liaw, *J. Electrochem. Soc.*, **163**, A2611 (2016).
53. M. Dubarry, M. Berecibar, A. Devie, D. Anseán, N. Omar, and I. Villarreal, *J. Power Sources*, **360**, 59 (2017).
54. M. Dubarry, C. Truchot, A. Devie, B. Y. Liaw, K. Gering, S. Sazhin, D. Jamison, and C. Michelbacher, *J. Electrochem. Soc.*, **162**, A1787 (2015).